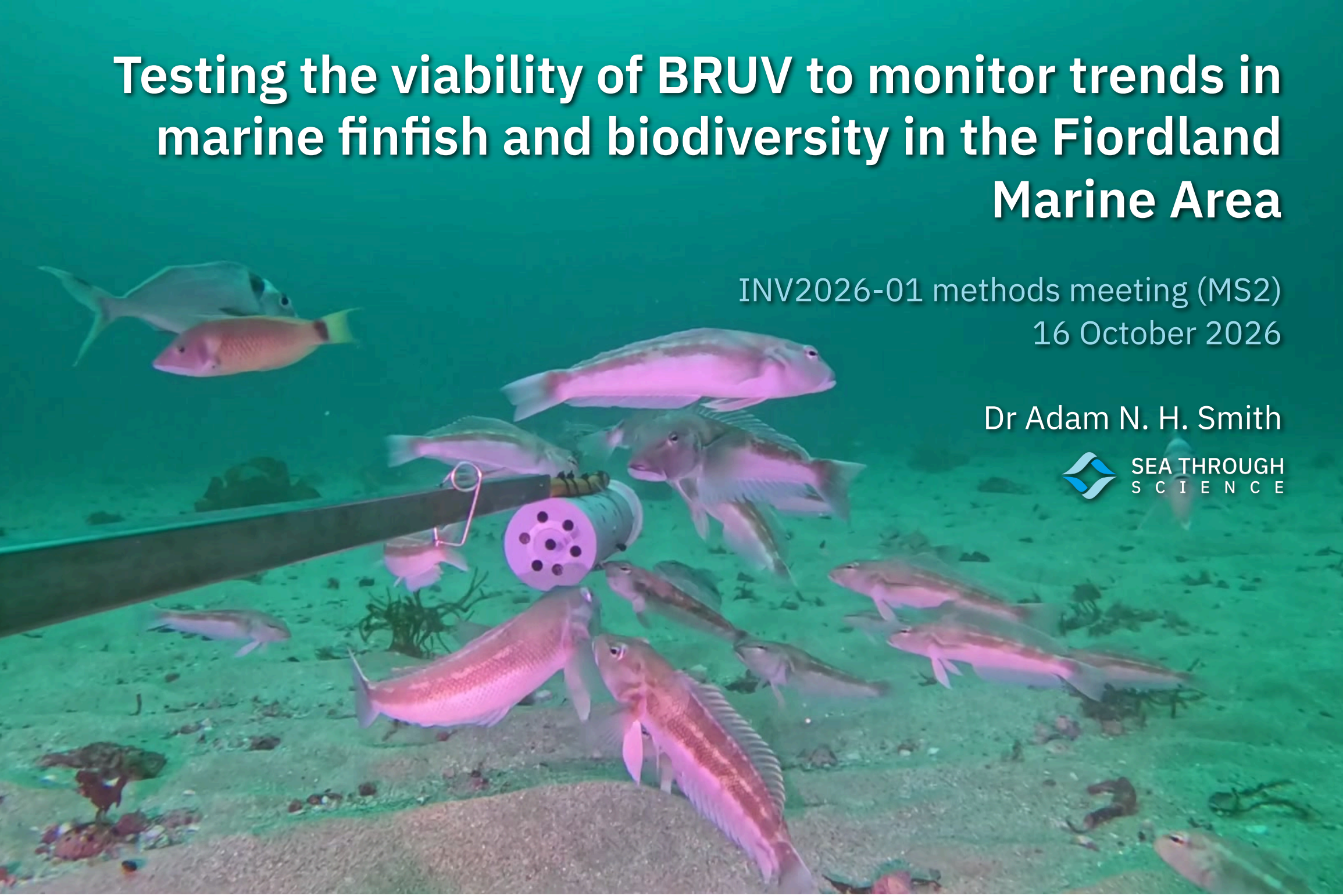
Testing the viability of BRUV to monitor trends in marine finfish and biodiversity in the Fiordland Marine Area

INV2026-01 methods meeting (MS2)
16 October 2026

Dr Adam N. H. Smith

SEA THROUGH SCIENCE



Objectives

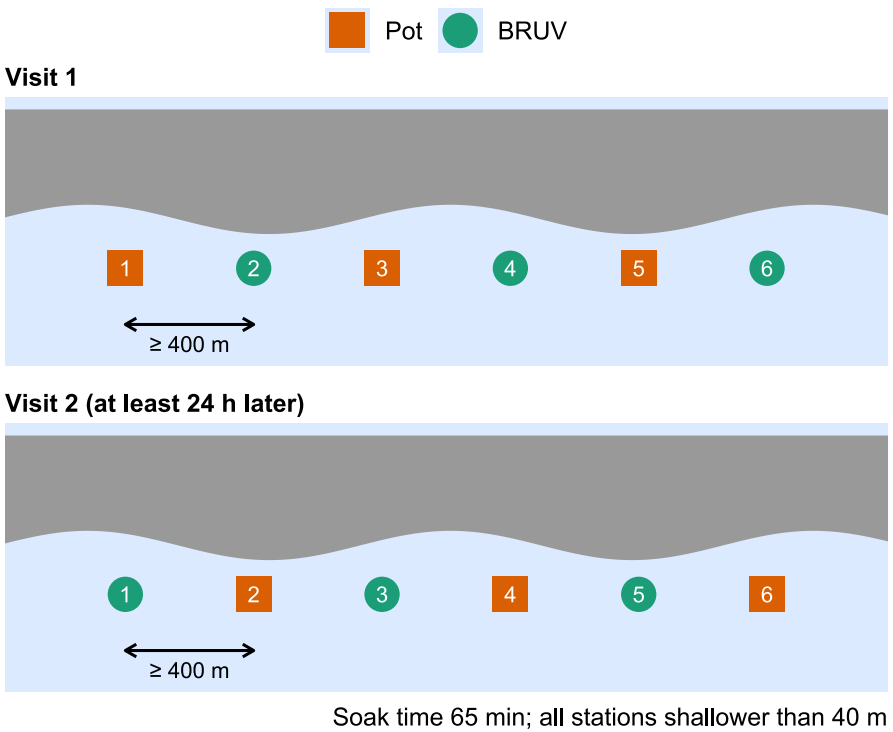
- **Objective 1:** annotation of stereo-BRUV footage from Dusky Sound: 102 deployments from 2025, and 94 deployments from 2024 updated to the current protocol
- **Objective 2:** field experiment comparing blue cod pots with stereo-BRUVs in Dusky Sound
- **Objective 3:** standardisation framework for Fiordland stereo-BRUV data, and the power of stereo-BRUV indices to detect change in fish populations

Milestones

Milestone	Deliverable	Due
MS3	Methods to FMG, BRAG and other working groups	30 Nov 2026
MS5	Annotation summary report and annotated dataset	30 Jun 2027
MS6	Draft AEBR; final results to BRAG	15 Dec 2027
MS7	Final AEBR	31 Mar 2028

Experimental design

Aim: compare catch rates and selectivity of pots vs BRUVs at the same stations — can the new BRUV monitoring be linked to the historical potting series?



Soak time 65 min; all stations shallower than 40 m

- Outer Dusky Sound: three strata, 36 stations, all shallower than 40 m
- Each station sampled by both gears in counterbalanced order (AB/BA crossover)
- 65 min soak from touchdown; FNZ potting manual where possible (Beentjes 2019)
- Stereo cameras on the pots if feasible (not essential to Objective 2)

Field timing

- Proposed window: late August to early September 2027
- Vessel: DOC *Southern Winds*, available July to September 2027

Pots: standardisation and camera mounting

- Use historical potting protocols from the FNZ potting manual as far as possible
- Counts and lengths only (no otoliths or sex determination)
- For discussion with Pete: pot design, bait, etc.
- Camera mounting options:
  1. inside the pot
  2. on the outside of the pot, facing the entrance
  3. on a separate frame facing the pot

Taumoana (Five Finger Peninsula) Marine Reserve

- Should the experiment include pot stations inside the reserve?
- Can potting inside the reserve be permitted?

Meetings

- FMG monitoring subcommittee: **11 November 2026**
  - Confirmed?
  - FMG-proper meeting?
- BRAG meeting: **24 November 2026** (MS3)

PDF of the slides circulated before each meeting

Planned content

- Approach to the whole project:
  - desktop project: standardisation framework for Fiordland stereo-BRUV data, with simulations of the power of stereo-BRUV indices to detect change
  - pot–BRUV field experiment
- Largely the same content for both presentations

Permits and approvals

- MPI special permit for potting — obtained for 2025 but not used
- DOC permit for any potting inside Taumoana (Five Finger Peninsula) Marine Reserve (if that’s the plan)
- Animal ethics approval from an ethics committee (Earth Sciences or Auckland Uni), because potting involves handling live fish
- Health and safety plan (MS4, due 31 May 2027)